

Environmental Impact Assessment of Maize Production and Study of 3 Potential Scenarios

Technical Description and Study Scenario Analysis (LCA)

To evaluate the environmental impacts of maize production, a Life Cycle Assessment (LCA) model was developed using **openLCA** software, utilizing the international scientific database **Agribalyse** and the **EF 3.1 Method** for impact assessment.

The functional unit was defined as **1 kg of maize at farm gate**, based on an average typical production yield of **10,000 kg per hectare** (1,000 kg/stremma).

Using dynamic local parameters (Input Parameters), three (3) different management scenarios were studied and compared:

Scenario 1: Reference Scenario

- **Description:** Represents the conventional, widely applied cultivation practice of the region, without the use of precision technologies or advanced resource-saving systems.
- **Input Data:** Application of **150 kg of Nitrogen (N) per hectare** and a total irrigation of **2,000 m³ of water per hectare**, along with typical quantities of phosphate fertilizers and diesel fuel for agricultural operations.
- **Environmental Footprint:** This scenario serves as the baseline ("ground zero") for comparison. It presents the highest carbon footprint at **0.07757 kg CO₂ eq per kg of product**, due to increased indirect emissions from fertilizer manufacturing, as well as direct nitrous oxide emissions into the air and nitrates into groundwater.

Scenario 2: Controlled Fertilization

- **Description:** Examines the implementation of precision agriculture technologies (e.g., soil/foilage sensors, variable rate maps), which allow for a rational reduction of nitrogen inputs without affecting the final crop yield.
- **Input Data:** A **26.6% reduction** in nitrogen input, setting the parameter at **110 kg N per hectare**. Other inputs (water, fuel) remain constant.
- **Environmental Footprint:** This scenario achieved the most significant environmental improvement, reducing the carbon footprint to **0.05722 kg CO₂ eq**, which is a decrease of approximately **19.4%** compared to the reference scenario. This reduction occurred because the openLCA model automatically calculated both the lower footprint from fertilizer production at the factory and the reduced direct greenhouse gas emissions in the field. Concurrently, the risk of freshwater eutrophication from nitrates is drastically reduced.

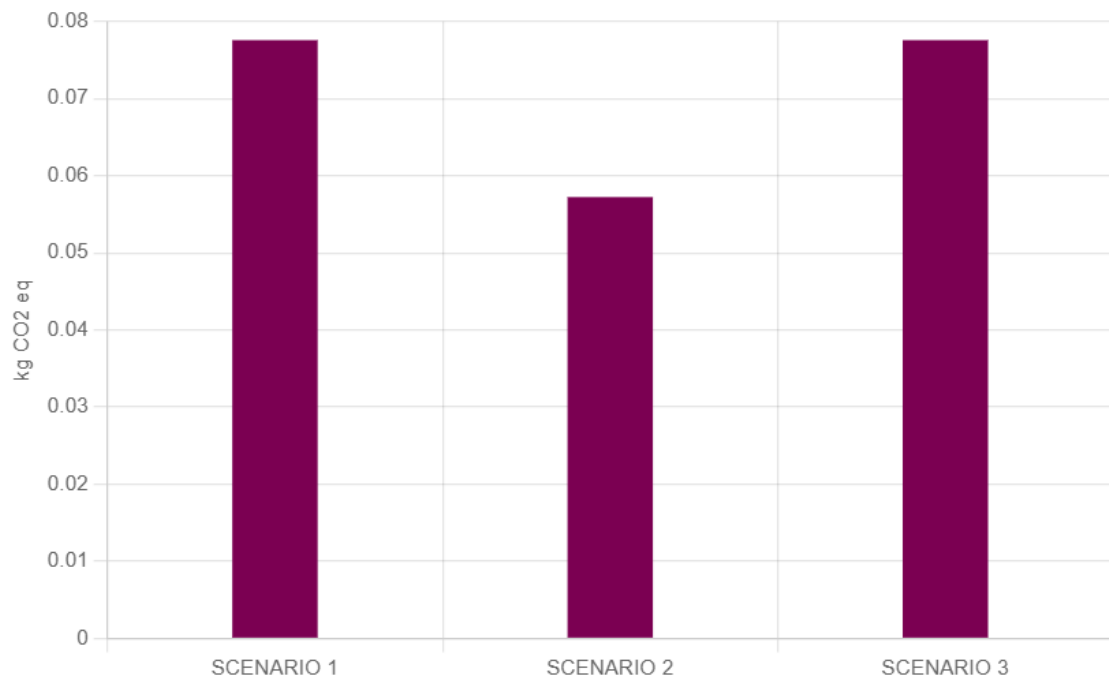
Scenario 3: Drip Irrigation Method

- **Description:** Examines the replacement of conventional, energy-intensive, and wasteful irrigation methods (e.g., hose reels or sprinklers) with a modern drip irrigation system to optimize water use.
- **Input Data:** Reduction in water consumption, setting the parameter at **1,200 m³ of water per hectare**. Fertilization remains at the reference scenario baseline (150 kg N).
- **Environmental Footprint:** In the Global Warming impact category, the footprint remains stable at **0.07757 kg CO2 eq**, as the consumption of natural water from the environment does not directly emit greenhouse gases. However, this scenario presents top-tier performance in the **Water Consumption** (Water Consumption / Resource Depletion) category, proving the importance of this technology for protecting water reserves and addressing water scarcity.

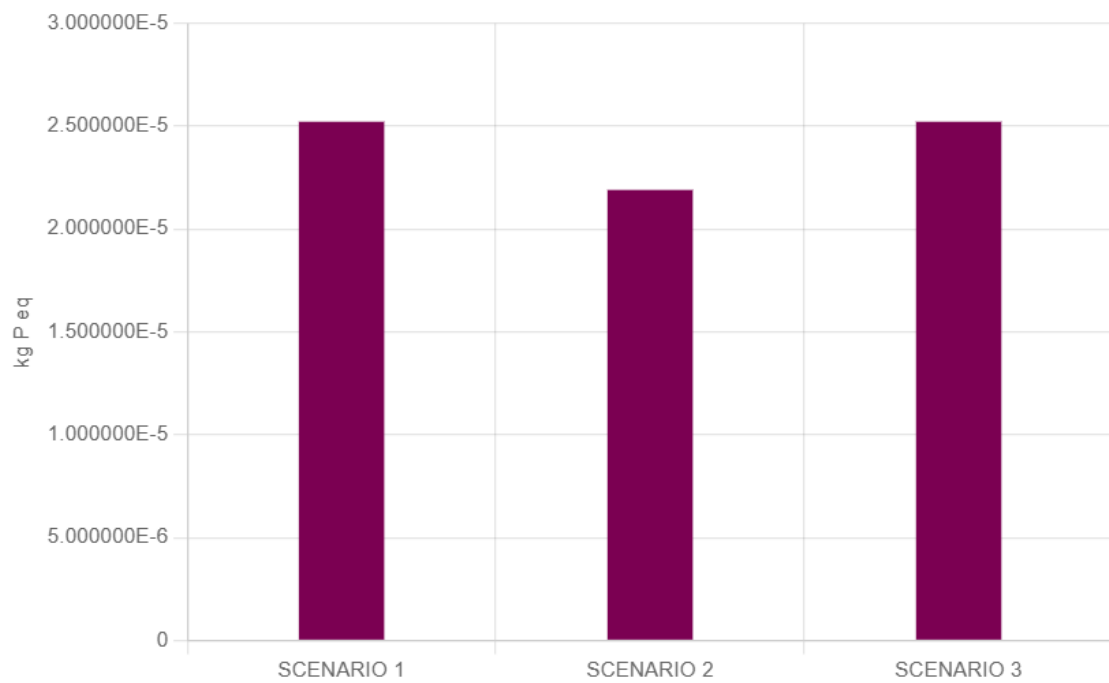
Analysis Results

Indicator	Reference Scenario	Controlled Fertilization Scenario	Drip Irrigation Scenario	Units
Freshwater eutrophication	2.52210E-5	2.19046E-5	2.52210E-5	kg P eq
Global warming	0.07757	0.05722	0.07757	kg CO2 eq
Water consumption	7.36526	7.35572	4.44286	m ³ world eq

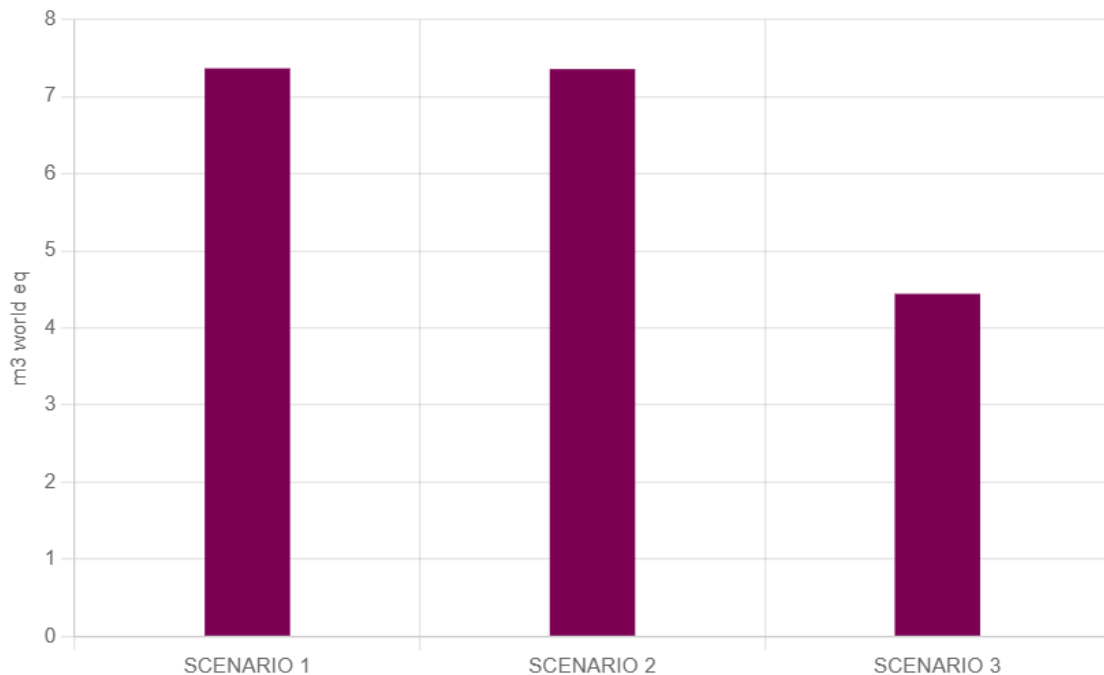
Global Warming Chart



Fresh water eutrophication chart



Water consumption chart



Key Research Findings

1. Global Warming

- **Scenario 2 (Controlled Fertilization):** Achieved the maximum reduction, as the rational use of nitrogen drastically limits direct nitrous oxide emissions from the soil, which is a potent greenhouse gas.
- **Scenarios 1 & 3:** Completely identical. Changing the irrigation system does not affect direct greenhouse gas emissions on the farm.

2. Freshwater Eutrophication

- **Scenario 2 (Controlled Fertilization):** Brought about a significant reduction. Decreasing fertilizer inputs limits the leaching and runoff of nutrients into aquatic ecosystems, preventing the phenomenon of eutrophication.
- **Scenarios 1 & 3:** Show no change, as eutrophication in this specific system is primarily determined by the availability of nitrogen and phosphorus.

3. Water Consumption

The reference scenario presents the maximum water consumption (**7.36526 m³**). In Scenario 2, the reduction in fertilization brings a marginal drop to **7.35572 m³**, limiting the indirect water used in the industrial synthesis of fertilizers. Conversely,

the application of precision systems in Scenario 3, such as drip irrigation, achieves the most drastic reduction to **4.44286 m³**. This differentiation confirms that direct control through technological modernization is the determining factor for reducing the total water footprint.

Conclusion

Achieving sustainable production requires a combined strategy, as no single practice covers all indicators. Controlled fertilization (Scenario 2) is crucial for mitigating Climate Change and Eutrophication, while drip irrigation (Scenario 3) constitutes the only effective solution for a drastic reduction in the water footprint.